

Name		Index No.		Civics Group	
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DUNMAN HIGH SCHOOL
SENIOR HIGH
 2008 PRELIMINARY EXAMINATIONS
 General Certificate of Education Advanced Level
 YEAR SIX H1 BIOLOGY (8875)

BIOLOGY

8875 / 01

PAPER 1 Multiple Choice

24 September 08 (Wednesday)

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in the **soft pencil** on separate Answer Sheet [**optical answer sheet (OTAS)**] provided.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

Do not open this booklet until you are told to do so.

This question paper consists of **15** printed pages and **1** blank page (including this cover page).

[Turn over]

Section A: Multiple Choice Questions (30 marks)

Answer all questions.

Refer to the following information from Questions 1 – 3.

To determine the intracellular pathways of three proteins **X**, **Y** and **Z**, four identical cultures of cells were labeled with ^3H -amino acids for 5 min.

One culture was harvested at the end of the 5 min labeling period. The other three cultures were rinsed and then were put into media with 20-fold excess of unlabelled amino acids for 10 min, 20 min and 30 min respectively before harvesting.

After harvesting, the cells were put on ice and broken open and the cellular fractions were obtained by differential centrifugation. The cellular fractions harvested included: mitochondria, rough endoplasmic reticulum (rough ER), Golgi apparatus, plasma membranes, free ribosomes and soluble cellular supernatant (cytosol).

In each of these fractions, the amount of newly synthesized protein was determined. The data, expressed as the percentage of the total newly synthesized protein of a particular type in each fraction, were recorded in the table below.

Harvest Time	5 min.			10 min.			20 min.			30 min.		
Protein	<i>X</i>	<i>Y</i>	<i>Z</i>	<i>X</i>	<i>Y</i>	<i>Z</i>	<i>X</i>	<i>Y</i>	<i>Z</i>	<i>X</i>	<i>Y</i>	<i>Z</i>
<u>Fraction</u>												
Mitochondria	0	0	0	0	0	0	0	0	50	0	0	100
Rough ER	0	100	0	0	75	0	0	10	0	0	0	0
Golgi apparatus	0	0	0	0	25	0	0	50	0	0	20	0
Plasma membrane	0	0	0	0	0	0	0	40	0	0	80	0
Ribosomes	100	0	100	25	0	0	0	0	0	0	0	0
Cytosol	0	0	0	75	0	100	100	0	50	100	0	0

- 1** The associations of proteins **X** and **Z** with free ribosomes at a harvest time of 5 min indicates which of the following about the proteins?

- A** They are ribosomal proteins.
- B** They are at the site of synthesis.
- C** They are glycosylated proteins.
- D** They are identical.

[Turn over]

2 The inferred steps in the pathway of protein Y are correctly designated by which of the following?

A Endoplasmic reticulum → Golgi apparatus → plasma membrane

B Endoplasmic reticulum $\begin{matrix} \nearrow \text{plasma membrane} \\ \searrow \text{Golgi apparatus} \end{matrix}$

C Plasma membrane → Golgi apparatus → endoplasmic reticulum

D Plasma membrane $\begin{matrix} \nearrow \text{endoplasmic reticulum} \\ \searrow \text{Golgi apparatus} \end{matrix}$

3 At a harvest time of 10 min, which of the following statements is correct about protein X compared with protein Z?

A Protein X is synthesised more rapidly than protein Z.

B Protein X is degraded more rapidly than protein Z.

C Protein X is released from ribosomes more slowly than protein Z.

D Protein X is released from ribosomes more rapidly than protein Z.

4 Collagen is a fibrous protein found in mammalian tendons. Which feature contributes most to the great tensile strength of collagen?

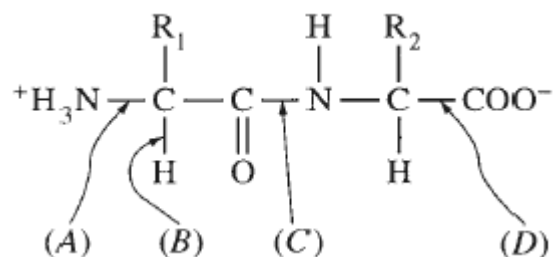
A a quaternary structure of triple helices bonded together with covalent and hydrogen bonds

B a secondary structure with many hydrogen bonds firmly holding α -helices

C a regularly folded tertiary structure held together with hydrogen bonds and ionic bonds

D a primary sequence with covalent bonds linking a variable sequence of amino acids.

- 5 A peptidase hydrolyses peptide bonds in small proteins. In the molecule shown below, which bond would be hydrolyzed?



- A A
 B B
 C C
 D D
- 6 In the presence of a fixed concentration of a competitive inhibitor, which of the following would best characterize an enzyme-catalyzed reaction when the concentration of the substrate is increased?
- A the inhibition does not change
 B the inhibition decreases
 C the K_M increases
 D the v_{max} decreases
- 7 During which of the following stages of the cell cycle will a diploid cell contain twice the amount of DNA found in a gamete?
- A prophase
 B entire S phase
 C entire G_1 phase
 D entire G_2 phase

- 8 The tRNA anticodons for the amino acids arginine, proline and leucine acid, respectively are GCU, GGA and GAA.

What are the corresponding DNA triplets for these amino acids?

	arginine	proline	leucine
A	CGA	CCU	CUU
B	CGA	CCT	CTT
C	GCA	GGA	GAA
D	GCT	GGA	GAA

- 9 The diagram shows part of the normal sequence of an mRNA molecule.

CCAUAGUGGUCCGUAAAAUGGC

A mutation in the DNA resulted in a polypeptide beginning with the following sequence.

Glycine – serine – proline – glycine - isoleucine

The DNA triplets for some amino acids are:

glycine	proline	leucine	serine	isoleucine
CGA	CCA	TTA	TCA	ATA
GGT	CCG	CTT	TCG	ATT
GGC		CTC		

Which mutation has occurred in the DNA molecule?

- A** an addition of an extra nucleotide
- B** a reversal in the order of nucleotides
- C** the loss of a nucleotide
- D** the replacement of one nucleotide by a different nucleotide

[Turn over]

- 10** One complete turn of the double helix of DNA contains 10 pairs of bases and is 3.4 nm long.

What is the approximate length of the DNA coding sequence of lysozyme, a protein of 129 amino acids?

- A** 44 nm
- B** 66 nm
- C** 113 nm
- D** 132 nm

- 11** The table shows the percentages of bases in DNA from various types of cell.

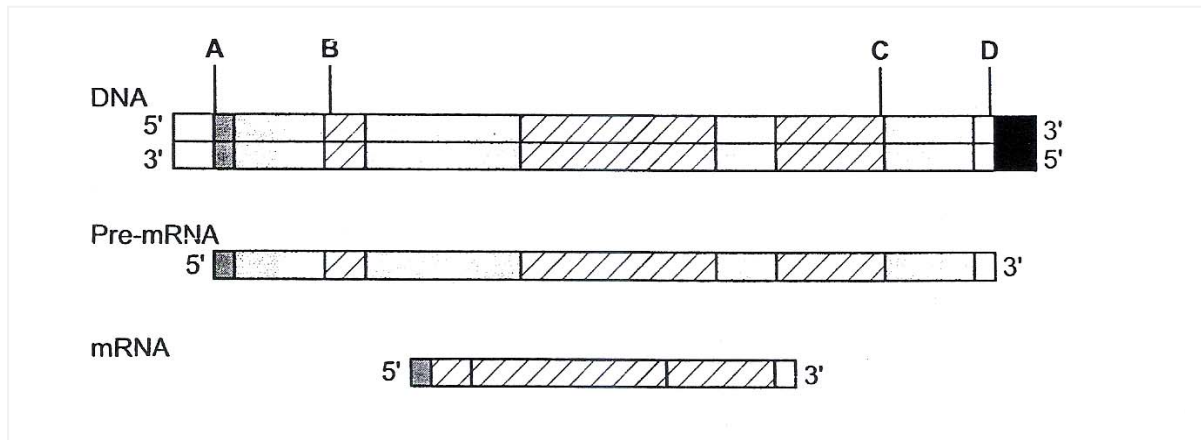
<i>source of DNA</i>	<i>adenine</i>	<i>guanine</i>	<i>thymine</i>	<i>cytosine</i>
ox thymus	28.2	21.5	27.8	22.5
ox spleen	27.9	22.7	27.3	22.1
ox sperm	28.7	22.2	27.2	22.0
rat bone marrow	28.6	21.4	28.4	21.5
yeast	31.3	18.7	32.9	17.1

Which of the following is a valid deduction from these data?

- A** DNA is composed of two strands twisted around each other.
- B** DNA occurs in about the same amounts in all cells from the same species.
- C** The four bases show complementary base pairing.
- D** The structure of DNA is the same in both yeast and animal cells.

- 12 The diagram shows the processing of transcribed pre-mRNA from a eukaryote.

At which part of the DNA molecule does transcription start?

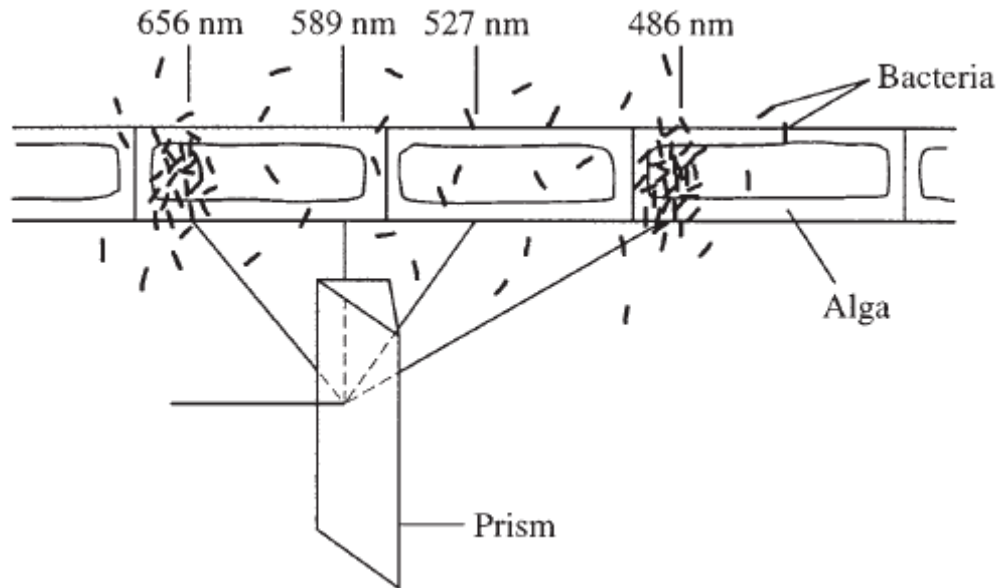


- 13 A region of eukaryotic DNA consists of over fifty repeats of the same sequence of twelve bases.

Where is this repetitive region *least* likely to be found?

- A an exon
 - B an intron
 - C a polymerase binding site
 - D satellite DNA
- 14 DNA fingerprints used as evidence in a murder trial look something like supermarket bar codes. The pattern of bars in a DNA fingerprint shows _____.
- A the order of bases in a particular gene
 - B the presence of various-sized fragments from chopped DNA
 - C the order of genes along particular chromosomes
 - D the exact location of a specific gene in a genomic library

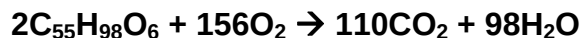
- 17 The filamentous alga *Cladophora* is illuminated with light dispersed by a prism. As shown in the diagram below, aerobic bacteria *Pseudomonas* included in the medium conglomerate where the alga is illuminated with light at 656 nm and 486 nm. *Pseudomonas* do not conglomerate in this manner if the *Cladophora* is removed from the medium.



Which of the following is the most likely explanation for the bacterial movement?

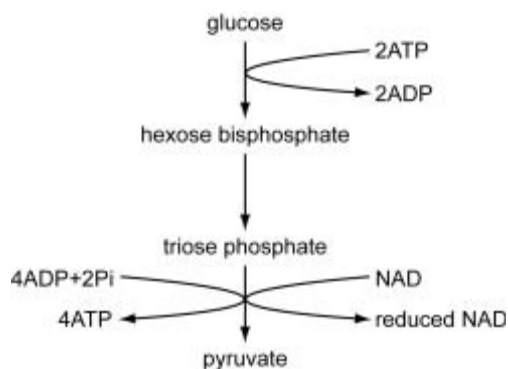
- A Starch made by photosynthesis is secreted from the alga in regions illuminated by light at 656 nm and 486 nm.
- B Alga and bacteria have a symbiotic relationship; the bacteria need oxygen supplied by the alga and the alga requires carbon dioxide supplied by the bacteria
- C Oxygen evolved by photosynthesis in the regions illuminated by light at 656 nm and 486 nm attracts bacteria.
- D Bacteria are repelled by high carbon dioxide content in regions illuminated by light at 589 nm and 527 nm.

- 18 The equation shows the complete aerobic respiration of a respiratory substrate:



The respiratory quotient (RQ) for respiration of this respiratory substrate is

- A 0.70
 B 0.71
 C 0.75
 D 1.42
- 19 The diagram summarizes glycolysis.



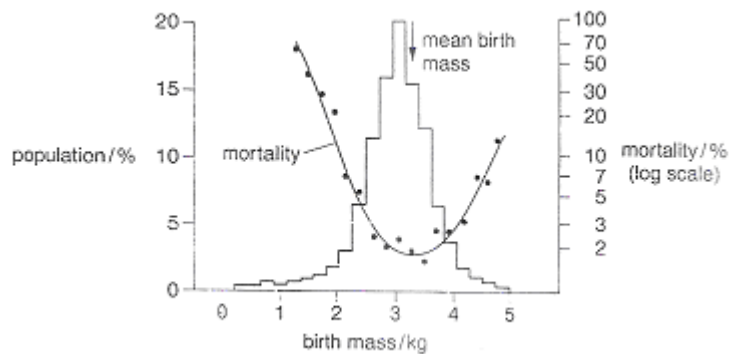
Which correctly gives the number of carbon atoms and phosphate groups in the four named molecules?

	atom/group	glucose	fructose biphosphate	glyceraldehyde phosphate	pyruvate
A	carbon	6	6	3	3
	phosphate	0	2	1	0
B	carbon	6	5	5	3
	phosphate	0	2	1	0
C	carbon	6	6	3	3
	phosphate	1	2	1	1
D	carbon	6	5	5	3
	phosphate	1	2	1	1

20 Mitochondria and chloroplasts carry out oxidative phosphorylation and photophosphorylation respectively by means of _____.

- A photorespiration
- B chemiosmotic coupling
- C high-energy intermediate coupling
- D conformational coupling

21 Various factors have influenced the evolution of humans.



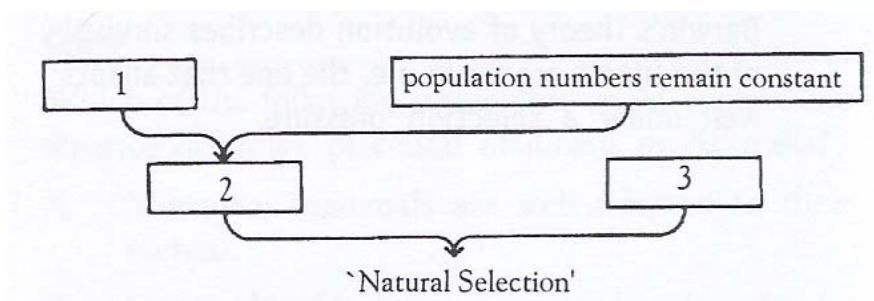
The graph illustrates one factor. Which factor does the graph illustrate?

- A Artificial selection
- B Directional selection
- C Disruptive selection
- D Stabilising selection

22 A large lake evaporates slowly over time, leaving many small ponds. The large lake had been inhabited by one species of fish, but over time, each small pond is inhabited by a different species of fish, each species a descendant of the species which inhabited the large lake. This is an example of _____.

- A punctuated equilibrium
- B disruptive selection
- C allopatric speciation
- D sympatric speciation

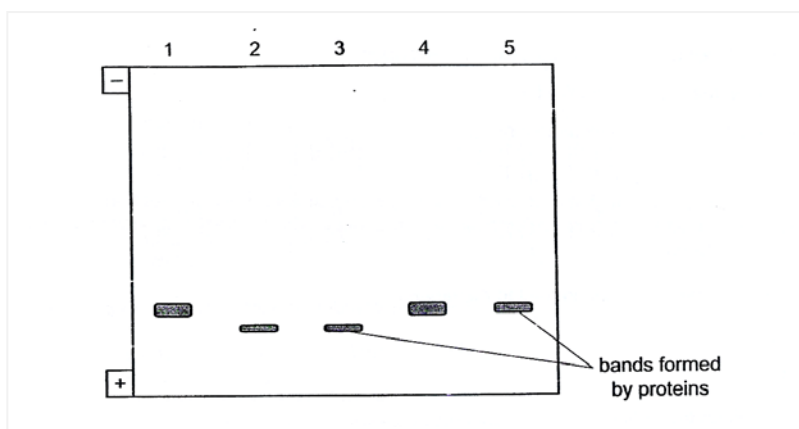
- 23 The diagram shows a summary of Darwin's theory of natural selection.



Which statements should be placed in boxes 1, 2 and 3?

	Individuals show great reproductive capacity	Variation is shown in all populations	There is a struggle for existence
A	1	2	3
B	1	3	2
C	2	3	1
D	3	1	2

- 24 The diagram shows the results of electrophoresis of the same protein extracted from five different individuals.



Which effect of mutation accounts for the different position of this protein extracted from individuals 2 and 3?

- A** A positively charged amino acid is being added to the protein.
- B** A non-charged amino acid replaces a negatively charged amino acid.
- C** A negatively charged amino acid replaces a non-charged amino acid.
- D** A negatively charged amino acid is being lost from the protein.

- 25 A researcher wants to clone a human gene whose primary RNA transcript contains introns. A bacterial plasmid is used as a vector to transfer the gene into bacteria. To ensure success of the procedure, all of the following steps should be included in the process **EXCEPT**:
- A Treat the plasmid and the human DNA with the same restriction enzyme.
 - B Use a restriction enzyme that does not produce sticky ends.
 - C Use the complementary DNA (cDNA) of the human gene.
 - D Provide suitable conditions (such as a calcium-rich growth medium) for transformation of the plasmid by the bacteria.
- 26 Five stages in the production of human insulin by genetic engineering techniques are given.
- 1 DNA cut with restriction enzymes
 - 2 DNA copy made using reverse transcriptase
 - 3 Messenger RNA extracted from cells
 - 4 Plasmid DNA joined to donor DNA using ligase enzymes
 - 5 Recombinant plasmid inserted into bacterial cell

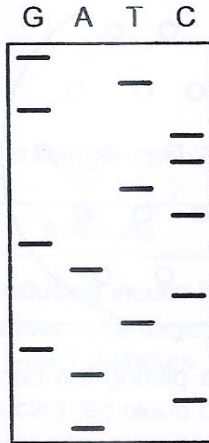
Which sequence is correct?

first —————> last

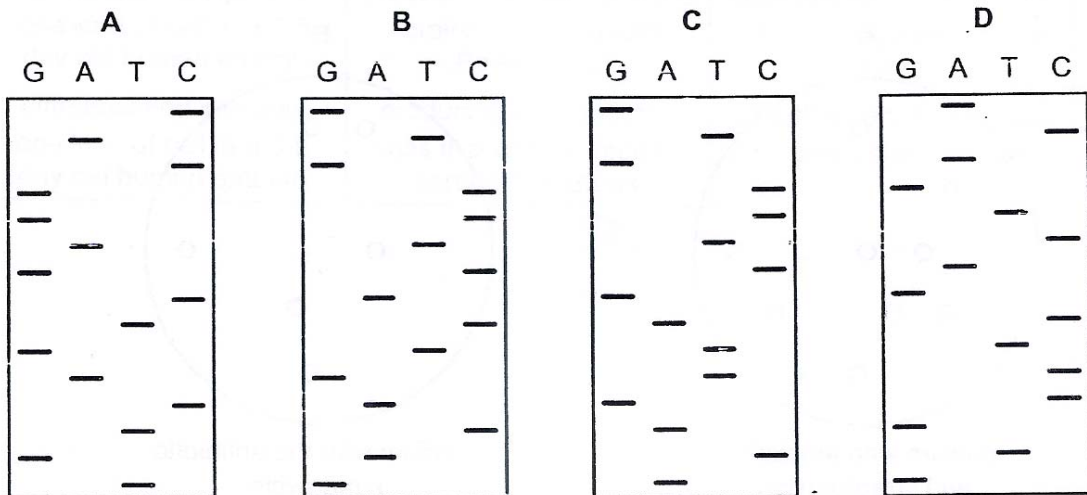
- A 1 → 3 → 2 → 5 → 4
- B 1 → 3 → 4 → 2 → 5
- C 2 → 1 → 3 → 5 → 4
- D 3 → 2 → 1 → 4 → 5

- 27 In people with a mutant allele, a protein contains just one different amino acid in its primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using electrophoresis.

The electrophoresis results from the DNA of a normal allele of a gene is shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?



28 Which feature of stem cells obtained from blood in the umbilical cord enables their use in the treatment of a variety of blood cancers?

- A** They can differentiate into bone marrow cells.
- B** They can differentiate into any cellular component of blood.
- C** They can replace blood stem cells affected by cancer.
- D** They are totipotent.

29 What are the normal functions of stem cells in a living human?

	Functions		
A	differentiating into many kinds of cells in 3 – 5 day old human embryo	producing insulin in pancreases damaged by type 1 diabetes	cells that can differentiate into bone cells in the skeleton
B	differentiating into many kinds of cells in 3 – 5 day old human embryo	producing red blood cells worn out by normal wear and tear	cells that can differentiate into cardiac muscle cells in the heart
C	differentiating into many kinds of cells in 3 – 5 day old human embryo	producing dopamine in the brains of people with Parkinson's	cells that can differentiate into cartilage cells in the joints
D	differentiating into many kinds of cells in 3 – 5 day old human embryo	producing differentiated cells that can be used to screen new drugs	cells that can differentiate into nerve cells in the brain

30 Synthesis of human insulin by genetically manipulated bacteria involves the use of the enzyme reverse transcriptase.

What is the role of this enzyme?

- A** It adds 'sticky ends' to insulin genes.
- B** It causes production of mRNA in the pancreas cells.
- C** It causes complementary DNA to be formed from mRNA.
- D** It causes single-stranded DNA to convert to double-stranded DNA.

END OF PAPER

Name		Index No.		Civics Group	
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DUNMAN HIGH SCHOOL SENIOR HIGH

2008 PRELIMINARY EXAMINATIONS

General Certificate of Education Advanced Level

YEAR SIX H1 BIOLOGY (8875)

BIOLOGY

8875 / 02

PAPER 2 Core Paper

12 September 08 (Friday)

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Do not open this booklet until you are told to do so.

Section B:

Consists of 7 Structured Questions

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section C:

Consists of 2 Essay Questions

Answer any **one** question.

Write your answers on the separate **writing papers** provided.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE		
Section A		
Section B	1	
	2	
	3	
	4	
	5	
	6	
	7	
Section C	8 / 9	
Total		

This question paper consists of **17** printed pages and **1** blank page (including this cover page).

Section B: Structured Questions (50 marks)

Answer **all** questions in this section.

Question 1

Table 1 shows the nucleotide sequences for some of the amino acids.

Table 1

CUU* leucine	CCU Proline	CAA glutamine	CGU arginine
AUU isoleucine	ACU threonine	AAU asparagine	GUU valine
GCU alanine	GAA glutamic acid	GGU glycine	UGG tryptophan

*messenger RNA codon corresponding to the amino acid

- (a) Using the information provided in **Table 1**, write out the base sequences that would produce the following chain of amino acids for

glycine – isoleucine – valine – glutamic acid – glutamine

in terms of

- (i) messenger RNA (mRNA). [1]

- (ii) DNA. [2]

- (b) The amino acids are assembled with the help of tRNA. Describe the structural features of a tRNA molecule which help it to adapt to its role in translation. [2]

Total: [5]

Question 2

The figure below shows a section of DNA, containing one gene.

	exon 1	intron	exon 2	intron	exon 3	
--	--------	--------	--------	--------	--------	--

P

- (a) (i) State **one** way in which the structure of this gene indicates that it is from a eukaryote. [1]

- (ii) Using the same diagrammatic format as in the figure, show what the pre-mRNA and mature mRNA will be for this gene. [3]

pre-mRNA:

mature mRNA:

- (b) Sometimes, a single nucleotide in the DNA sequence is replaced by a different nucleotide. Explain the possible consequence of a substitution at position **P** on the structure of the resulting protein. [2]

Total: [6]

Question 3

Different inbred lines of chickens show pronounced differences in mortality following infection with equal doses of two different strains of the bacterium *Salmonella typhimurium*.

In an investigation into the inheritance of resistance, a series of crosses was performed between two inbred lines of White Leghorn chickens, one line resistant to infection by the bacterium and one susceptible. The results of infecting newly hatched chicks of each inbred line, and of the cross between a susceptible male and a resistant female, are shown in **Table 3**.

Table 3

line of chickens	mortality / % of chickens infected	
	strain of <i>S. typhimurium</i>	
	F98	Swindon
resistant	40.0	0.0
susceptible	100.0	35.7
cross between susceptible ♀ and resistant ♂ (F ₁ hybrid)	45.0	0.0

It is thought that resistance to *S. typhimurium* is controlled by a single gene.

- (a) (i) Using suitable symbols, draw a genetic cross to show how resistance to *S. typhimurium* could be inherited in the cross that produces the F₁ hybrid shown at the bottom of **Table 3**. [3]

Working for genetic cross

Working for genetic cross (cont'd)

- (ii) Predict the ratio of resistant to susceptible chicks resulting from crosses between the following birds: [3]

Susceptible x F₁ hybrid

Resistant x F₁ hybrid

F₁ hybrid x F₁ hybrid

Total: [6]

[Turn over]

Question 4

2,4-dinitrophenol is a chemical that is toxic to mitochondria. When added to mitochondria this chemical allows electron transport to occur but prevents the phosphorylation of ADP to ATP. The chemical achieves this by breaking the essential link between electron transport and ATP synthesis. This toxin causes mitochondria to produce heat instead of ATP. The greater the amount of toxin added, the quicker is its action.

- (a) If mitochondria are poisoned with 2,4-dinitrophenol by what process could a plant cell produce more ATP? [1]

- (b) A researcher wanted to study cellular respiration in insect cells. She cultured some muscle cells from the common field cricket, *Teleogryllus oceanicus*, and then studied the effects of adding 2,4-dinitrophenol to these cells. An agricultural company may want to fund this research.

Suggest why an agricultural company might want to fund research on the effects of this toxin on field crickets. [1]

The experiment is summarized in **Table 4** below. Temperature observations in each trial were made at equal time intervals.

Table 4

Observations made at equal time intervals	Temperature / °C		
	Control (no 2,4- dinitrophenol)	Trial 1 (2,4- dinitrophenol added)	Trial 2
1 (start)	28	28	28
2	27	28	29
3	28	29	
4	29	31	
5	28	36	
6	28	23	
7	27	21	

- (c) Trial 2 had twice the concentration of 2,4-dinitrophenol added.

Complete the table by writing in temperatures in the spaces provided to predict the trend. [2]

Another researcher suggested adding pyruvate to the cells at the beginning of the experiment to cancel out the effects of this toxin.

- (d) Explain what effect adding pyruvate would have on cancelling out the effect of this toxin. [2]

Total: [6]

Question 5

- (a) The bird's nest fern, *Alocasia macrorrhiza*, usually grows in deeply shaded rainforests and has dark green fronds (leaves). Sometimes it is found in open, sunny locations by roadsides where it tends to have lighter colored fronds.

Two bird's nest ferns, one from each of the two habitats described above, were examined. A sample of cells from a frond of each of the ferns was collected. These cells were examined under an electron microscope and a typical chloroplast from each habitat (deeply shaded rainforest and sunny location) was drawn. These drawings are shown below in **Figure 5.1**.

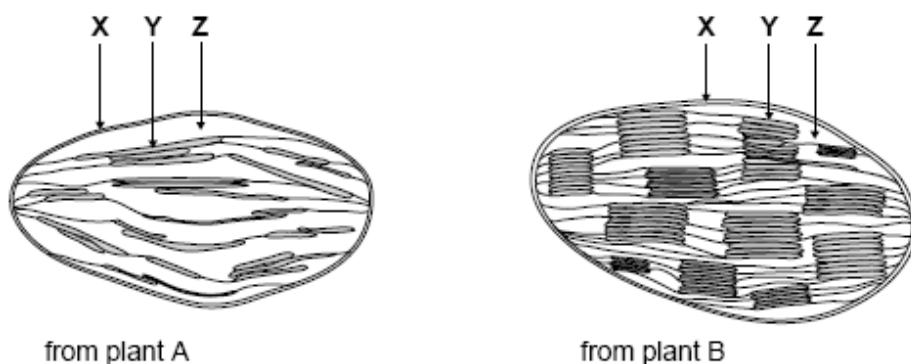


Figure 5.1

- (i) Which of the labeled parts, X, Y or Z, absorb light energy? **Name** the part. [1]

- (ii) Which drawing, **A** or **B**, shows a chloroplast from the rainforest habitat? Explain the reason for your choice in terms of the relationship between structure and function. [2]

- (b) It is now widely accepted by biologists that chloroplasts and mitochondria were once independent prokaryotic organisms which came to live symbiotically inside larger eukaryotic cells. This idea is known as the endosymbiotic theory.

Name two structural features of chloroplasts or mitochondria and outline how they support the endosymbiotic theory for the origin of these organelles. [2]

- (c) The simplified diagram **Figure 5.2** below summarizes the light-independent reactions (Calvin cycle) of photosynthesis.

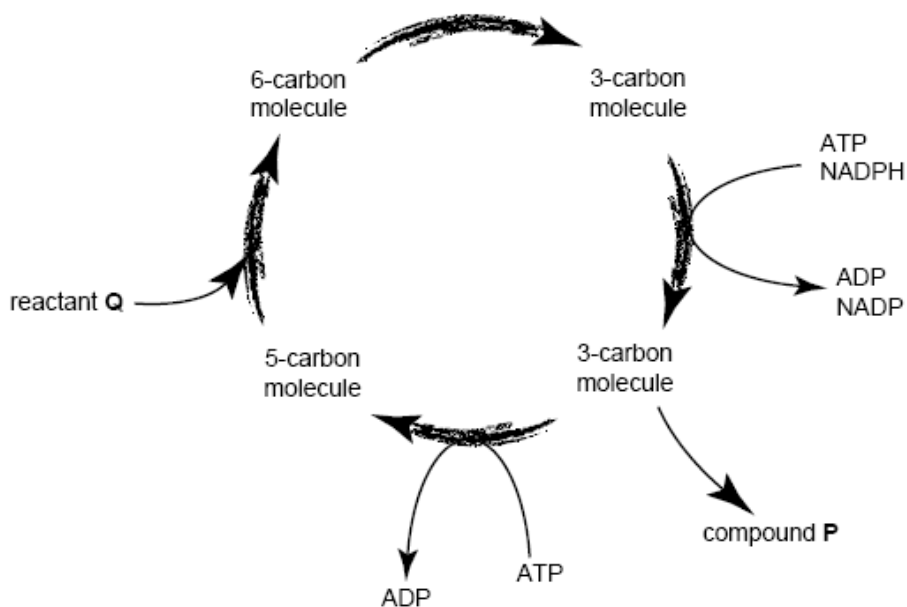


Figure 5.2

Name compound **P** and reactant **Q**. [2]

P:

Q:

Total: [7]

Question 6

(a) Figure 6 shows the main stages in the polymerase chain reaction (PCR).

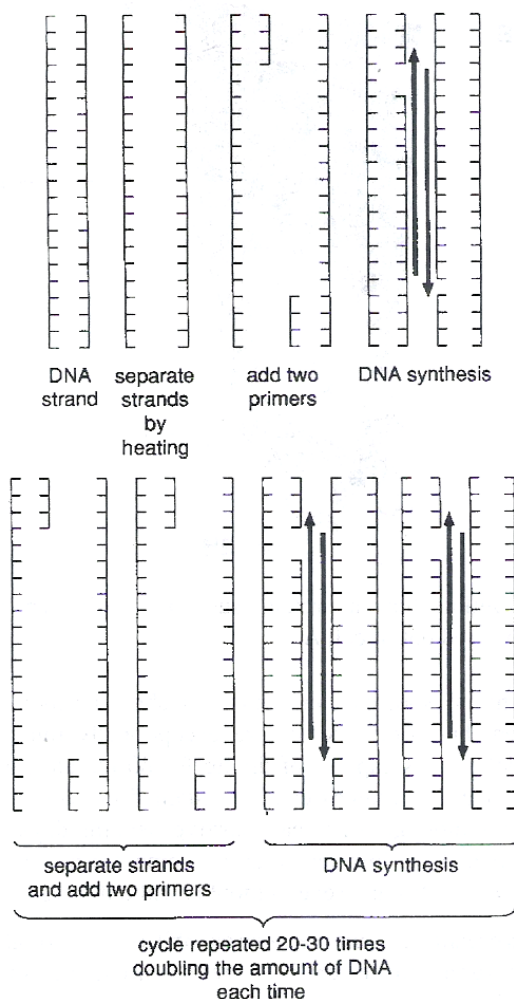


Figure 6

This technique allows multiple copies of a selected length of DNA to be made in the laboratory. Portions of the base sequences at either end of the DNA to be copied are used to produce two short lengths of synthetic DNA, one complementary to each strand of the DNA double helix. These short lengths of DNA serve as primers for *in vitro* DNA synthesis, which is catalysed by DNA polymerase, and they determine the ends of the final DNA fragment that is obtained.

Each cycle of the reaction requires a brief heat treatment to separate the two strands of the DNA double helix. A subsequent cooling of the DNA in the presence of a large excess of the two DNA primers allows their specific

attachment to their complementary DNA polymerase and the four different DNA nucleotides so that the regions of DNA, starting from the primers, are selectively synthesised.

When first developed, PCR was slow and laborious because a crucial enzyme for the process, DNA polymerase was denatured at high temperature and had to be added to each cycle. However the introduction of the thermostable enzyme, *Taq* DNA polymerase (isolated from the heat tolerant bacterium, *Thermus aquaticus*) revolutionised PCR so that the technique could be fully automated.

- (a)** (i) List the components of a nucleotide. [2]

- (ii) Explain what is meant by *four different nucleotides*. [2]

- (b)** Suggest why the brief heat treatment separates the two strands of the DNA double helix. [2]

- (c) Suggest why DNA synthesis takes place in opposite directions on the two separated strands of the double helix. [2]

- (d) Explain the advantages of using *Taq* DNA polymerase. [2]

Total: [10]

Question 7

A possible route for the escape of an inserted gene from a genetically engineered crop of oil-seed rape, *Brassica napus*, is into populations the wild turnip, *Brassica rapa*.

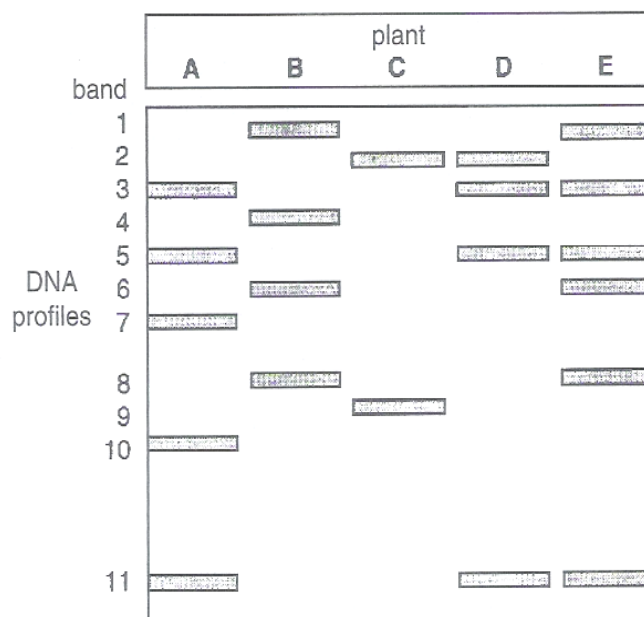
- (a) Explain briefly why there is a possibility of an inserted gene escaping from oil-seed rape to wild turnip. [2]

- (b) Outline the use in DNA analysis of

- (i) restriction enzymes; [2]

- (ii) electrophoresis; [2]

Some of the bands of the DNA profiles of the oil-seed rape cultivar (A), two wild turnip plants (B and C) and two possible interspecific hybrids (D and E) are shown in **Figure 7**.



key
A oil-seed rape
B wild turnip plant 1
C wild turnip plant 2
D possible interspecific hybrid
E possible interspecific hybrid

Figure 7

(c) State, giving reasons,

(i) Whether plants **D** and **E** are interspecific hybrids; [2]

(ii) From which wild turnip plant each was derived. [2]

D

E

Total: [10]

[Turn over]

Section C: Free-response Questions (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, wherever appropriate.

Your answer must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 **(a)** Describe the structure of the cell membrane and relate it to its functions. [14]
- (b)** The phospholipid bilayer forms an essential part of living cells but can also serve as a hindrance to certain cellular processes. Highlight two distinct examples and explain how the cell copes with this problem. [6]
- 9 **(a)** Explain how natural selection may bring about evolution. [7]
- (b)** Discuss the sources of variation important for natural selection to occur. [7]
- (c)** Describe **one** example of artificial selection. [6]

Total: [20]

END OF PAPER

[Turn over]